

Proposed Solution

Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	

Proposed Solution:

Opti-Crop is a smart agricultural decision support system that leverages machine learning and data analytics to assist farmers in making informed crop management decisions. The system analyzes agricultural data such as soil characteristics, weather conditions, rainfall, humidity, temperature, and historical crop production to provide accurate crop recommendations and yield predictions.

The application preprocesses the input data, applies trained machine learning models, and identifies meaningful patterns in agricultural production. Based on these insights, it recommends the most suitable crops, predicts expected yields, and suggests efficient utilization of resources such as water and fertilizers. By integrating environmental analysis and predictive analytics, OptiCrop helps farmers improve productivity, reduce cultivation risks, and support sustainable farming practices.

Functional Modules:

Module	Function
User Interface	Collects agricultural input from users
Data Collection Module	Loads agricultural datasets
Data Preprocessing Module	Cleans and prepares data
Machine Learning Module	Generates predictions
Crop Recommendation Module	Suggests suitable crops
Yield Prediction Module	Predicts crop production
Pattern Analysis Module	Identifies agricultural trends
Environmental Analysis Module	Evaluates climate and soil conditions
Resource Optimization Module	Suggests efficient use of resources
Reporting Module	Displays graphs and downloadable reports

Technologies Used:

Layer	Technology
Frontend	HTML, CSS
Backend	Flask (Python)
Database	SQLite
Machine Learning	Scikit-learn
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Model Storage	Joblib/Pickle
Version Control	Git & GitHub